

Reminder Scheduling Logic Design

Overview

This component determines when users should receive reminders to complete required forms, enter health data, or complete pending tasks. It evaluates user activity and scheduled reminder settings to decide whether a reminder should be generated. The scheduling logic is responsible only for determining eligibility and timing; notification delivery is handled by the notification system.

Reminder Types

The system supports the following reminder types:

- **Daily Reminder** – prompts users to complete required forms or data entry for the current day
- **Weekly Reminder** – prompts users to complete required forms or data entry for the current week
- **Incomplete To-Do Reminder** – prompts users to complete pending required actions such as unsubmitted forms or missing data entries

Scheduling Rules

A reminder is generated only when all of the following conditions are met:

- The reminder setting is active
- The scheduled execution time has been reached ($\text{next_run_at} \leq \text{current time}$)
- The required action is incomplete for the relevant period
- No duplicate reminder exists for the same user, reminder type, and time period

If the required action has already been completed for the relevant period, no reminder is generated and the system updates the next scheduled run time.

Time-Based Logic

The system determines completion based on the reminder type:

- **Daily Reminder** → Check if the user has submitted required data **today**
- **Weekly Reminder** → Check if the user has submitted required data **during the current week**
- **Incomplete To-Do Reminder** → Check if the required item is still incomplete or overdue

Duplicate Prevention

To avoid excessive or repeated notifications, the system enforces the following rules:

- Only one daily reminder per user per day
- Only one weekly reminder per user per week
- Only one reminder per incomplete to-do item within the same reminder period

Scheduling Process

The scheduling logic follows this process:

1. Load all active reminder settings
2. Select reminders where next_run_at ≤ current time
3. Check the user's completion status for the relevant period
4. If the action is incomplete → generate a reminder event
5. If the action is complete → skip reminder generation
6. Update next_run_at to the next scheduled time

Assumptions

- Required forms and missing data entries are treated as to-do items
- The scheduler runs at regular intervals (e.g., hourly)
- Reminder generation is separate from notification delivery